Exercise 4: Matrices and waves

Prognostic Modelling (pea700), WiSe 2026/27

Out: week 4 · Back: week 5 · 10 points

Covers unit 4 of the script. Start in the lab and finish at home.

Problem 1: A model matrix and its eigenvalues (4 points)

a) Set up the $N \times N$ model matrix A with 2 on the main diagonal and -1 on the two neighbouring diagonals for $N = 100$. Compute its eigenvalues and eigenvectors and compare the eigenvalues with $\lambda_m = 2 - 2\cos(m\pi/(N+1))$, $m = 1, \ldots, N$. Plot the eigenvectors belonging to the two smallest and the largest eigenvalue.

b) Compute the condition number of A for $N = 10, 100, 1000$. How does it depend on N? Explain the dependence with the formula for λ_m.

Problem 2: Layered ground (3 points)

Heat conduction through layered ground, $-(Ku')' = f$, has a conductivity K that changes from layer to layer. With $K_{j+1/2}$ the conductivity between grid points j and $j+1$, row j of the discretization reads

$$-K_{j-1/2}\,u_{j-1} + (K_{j-1/2} + K_{j+1/2})\,u_j - K_{j+1/2}\,u_{j+1} = h^2 f_j,$$

with Dirichlet values $u_0 = u_{N+1} = 0$. For $K = 1$ this is the model matrix A.

a) Write down the 3×3 matrix A_K for $N = 3$, with the four conductivities $K_{1/2}, K_{3/2}, K_{5/2}, K_{7/2}$. Show that

$$\vec{x}^T A_K \vec{x} = K_{1/2}\,x_1^2 + K_{3/2}(x_1 - x_2)^2 + K_{5/2}(x_2 - x_3)^2 + K_{7/2}\,x_3^2$$

and conclude that A_K is positive definite if all conductivities are positive.

b) Is A_K still positive definite (i) if $K_{5/2} = 0$, an insulating layer inside the domain, and (ii) if $K_{1/2} = K_{7/2} = 0$, no heat flow across either boundary? If not, give a vector $\vec{x} \neq \vec{0}$ with $\vec{x}^T A_K \vec{x} = 0$. Check your answers with the eigenvalues for conductivities of 1 and 0.

Problem 3: A spectrum and aliasing (3 points)

The signal

$$u(x) = \sin(2\pi \cdot 3x) + 0.5 \sin(2\pi \cdot 20x)$$

on $0 \leq x < 1$ contains 3 and 20 waves in the domain. It is sampled at N points $x_j = j/N$, $j = 0, \ldots, N-1$.

a) For $N = 64$, compute the amplitude spectrum with `np.fft.rfft` as in the script (amplitude $2|\hat{u}_\ell|$ for ℓ waves in the domain) and plot it. Which waves do you find, and with which amplitudes?

b) Repeat for $N = 32$. Where does the short wave appear now? Explain the number you find, and plot the samples together with the two candidate waves. From which N on is the spectrum correct?